

# MUHAMMAD UMER KHAN

AI Engineer | Agentic AI & RAG | Voice AI & Workflow Automation

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## SUMMARY

AI Engineer with 2+ years building production LLM systems — agentic RAG assistants, multi-agent backends, voice agents, and CRM automation. Delivered 12+ projects end to end — retrieval, fine-tuning, evaluation-gated CI deploys — including white-label SaaS live in production. Red-teams own systems against OWASP LLM Top 10; findings published.

## TECHNICAL SKILLS

**Agentic AI & LLMs:** RAG, Multi-Agent Systems, Prompt Engineering, LangChain, LangGraph, CrewAI, Hugging Face  
**Fine-Tuning & LLM Evaluation:** LoRA/PEFT, 4-bit Quantization, DeepEval, RAGAS, LLM-as-Judge, LangSmith, Logfire  
**Retrieval & Vector Databases:** Qdrant, FAISS, Hybrid Search (BM25 + dense), Rank Fusion, FlashRank Reranking  
**LLM Safety:** NeMo Guardrails, Llama Prompt Guard, Prompt-Injection Defense, OWASP LLM Top 10, MITRE ATLAS  
**MLOps & Deployment:** FastAPI, REST APIs, Docker, GitHub Actions CI/CD, MLflow, Streamlit, Portkey, OpenRouter  
**ML & Data Science:** Python, PyTorch, TensorFlow, Scikit-Learn, NLP, Computer Vision, Pandas, NumPy, SQL  
**Voice & Automation:** Vapi, Retell AI, Whisper ASR, n8n, Make, HubSpot, GoHighLevel, Airtable  
**Full-Stack & Cloud:** React, Next.js, Expo (React Native), Supabase, PostgreSQL, MongoDB, Git, AWS, Azure  
**Languages:** English (Professional), Urdu (Native)

## EXPERIENCE

### AI Engineer — Revera Innovations

Karachi (Hybrid) | Feb 2025 – Present

- Designed and deployed to production a virtual try-on system and an autonomous skin-analysis agent that detects skin concerns from a user's photo and recommends products from live inventory through agent reasoning.
- Shipped two white-label society-management SaaS platforms (React admin panels, Expo mobile apps, Supabase backends) — one live in production with offline HMAC-signed QR gate security and multi-tenant isolation.
- Automated lead capture and CRM operations with n8n and Make workflows across HubSpot, GoHighLevel, Airtable, and Jotform; delivered real-time AI voice agents on Vapi and Retell AI.

### AI Developer — Bright Solutions

Karachi (Hybrid) | Dec 2024 – Feb 2025

- Launched AI voice agents and automated lead-generation systems adopted in production for daily operations.

### AI Engineer Intern — SaylaniTech

Karachi (On-site) | Jun 2024 – Dec 2024

- Developed LLM- and NLP-based AI features and integrated them into applications through production-grade APIs.

### Data Science Intern — Digital Empowerment Network

Remote | May 2024 – Jun 2024

- Built predictive models on social-impact datasets; delivered data-cleaning pipelines and reports in Python.

## PROJECTS

### AskUmer — Hybrid-RAG Conversational Assistant

[GitHub](#) | 2026

- Engineered a LangGraph ReAct agent over hybrid retrieval — Qdrant dense + BM25 sparse with rank fusion and FlashRank reranking across 898 indexed chunks — plus an in-memory knowledge graph for multi-hop queries.
- Hardened with NeMo Guardrails input gating and Portkey failover (gpt-oss-120b → gpt-oss-20b, retries, caching); a DeepEval LLM-as-judge CI suite gates deploys at ≥ 0.7 faithfulness and answer relevancy.

### DineMate — Agentic AI Food-Ordering System

[Live Demo](#) · [Security Report](#) | 2025

- Built a four-agent LangGraph backend (ordering, order management, kitchen, admin) with voice ordering via Whisper ASR and TTS, role-gated dashboards, and analytics on revenue, peak hours, and top items.
- Red-teamed it against OWASP LLM Top 10 and MITRE ATLAS across 8 attack categories; fixed a critical price-manipulation flaw via server-side recomputation and deployed a Llama Prompt Guard 2 input classifier.

### SupportGenie — Dual Fine-Tuned Banking Support Chatbot

[GitHub](#) | 2025

- LoRA fine-tuned Mistral-7B (17,000 instruction pairs, r=32) and LLaMA-3-8B (1,764 pairs) in 4-bit quantization on a free-tier T4 to final loss 1.78 / 1.49, with FAISS FAQ retrieval falling through to the fine-tuned models.

### Diagnosisfy — LLM-Powered Medical Report Insights

[Live Demo](#) | 2025

- Top-10 finalist at the SMIT AI Hackathon, built in 12 hours: an OCR + LLM pipeline classifying lab values Normal / Borderline / Critical, with follow-up Q&A through a FAISS-grounded RAG chatbot scored with RAGAS.

## EDUCATION & CERTIFICATIONS

### B.S. in Computer Science — Sindh Madressatul Islam University

Karachi | 2023 – 2027 (Expected)

- Deep Learning Specialization · Natural Language Processing Specialization — DeepLearning.AI (Coursera), 2025
- Machine Learning Professional Certificate — IBM (Coursera), 2024 · AI & Data Science — SMIT (Saylani), 2025